

DBS LAB 3

IN – LAB TASK

TASK 1:

The screenshot displays the Oracle SQL Developer environment. The top toolbar includes icons for running, saving, and editing scripts. The 'Query Builder' tab is active, showing a SQL script to create a table named 'Employees2'. The script defines five columns: 'Employee_ID' as a primary key, 'Employee_Name' as a non-null varchar, 'Hire_Date' as a date, 'Job_Title' as a varchar, and 'Salary' as a number. The bottom 'Script Output' pane shows the successful execution of the script, with a message 'Table EMPLOYEES2 created.' and a completion time of 0.103 seconds.

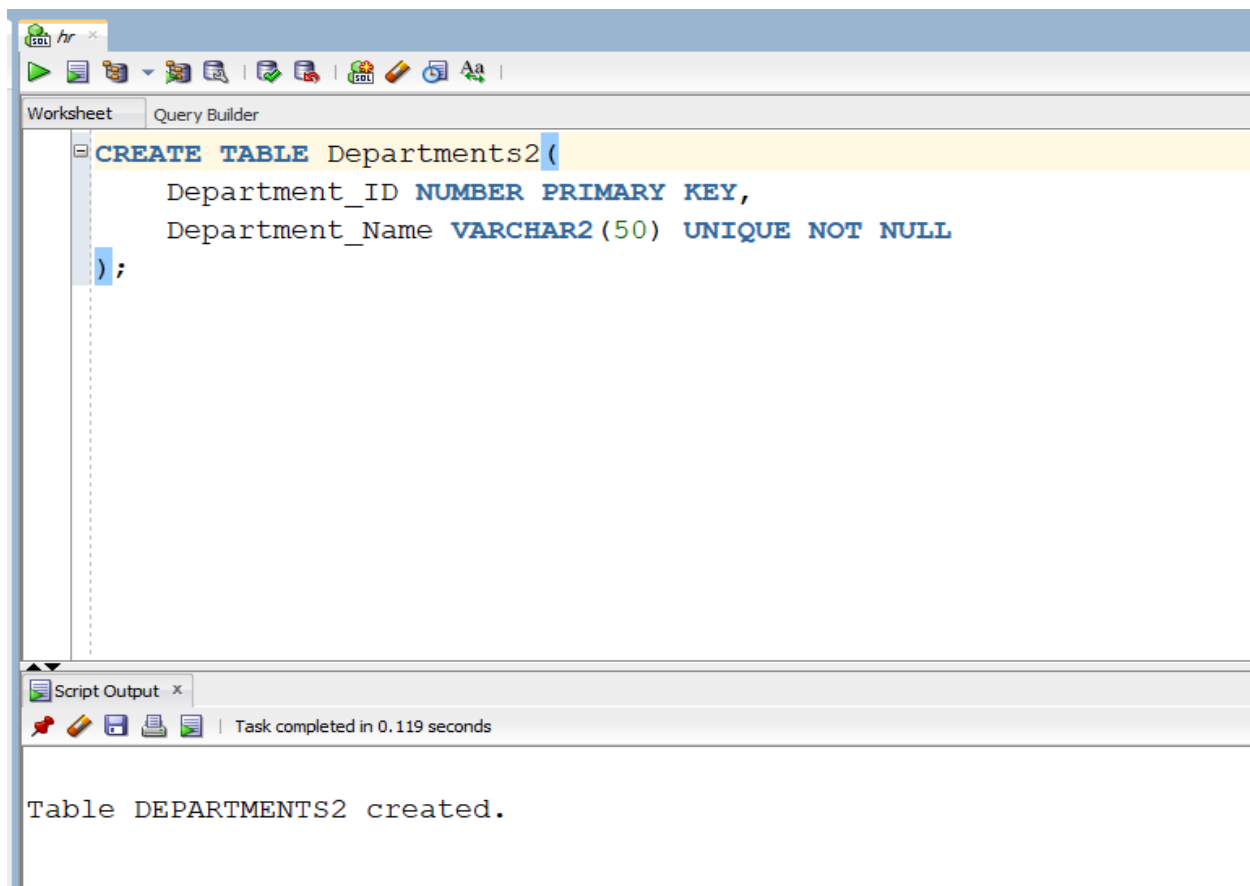
```
CREATE TABLE Employees2(  
    Employee_ID NUMBER PRIMARY KEY,  
    Employee_Name VARCHAR2(50) NOT NULL,  
    Hire_Date DATE,  
    Job_Title VARCHAR2(50),  
    Salary NUMBER(10,2)  
);
```

Script Output x

Task completed in 0.103 seconds

Table EMPLOYEES2 created.
|

TASK 2:



The screenshot displays the Oracle SQL Developer environment. The main window is titled 'hr' and shows a 'Query Builder' tab. The SQL statement entered is:

```
CREATE TABLE Departments2(  
    Department_ID NUMBER PRIMARY KEY,  
    Department_Name VARCHAR2(50) UNIQUE NOT NULL  
);
```

Below the SQL editor, the 'Script Output' window shows the result of the execution: 'Task completed in 0.119 seconds'. Below the screenshot, the text 'Table DEPARTMENTS2 created.' is displayed.

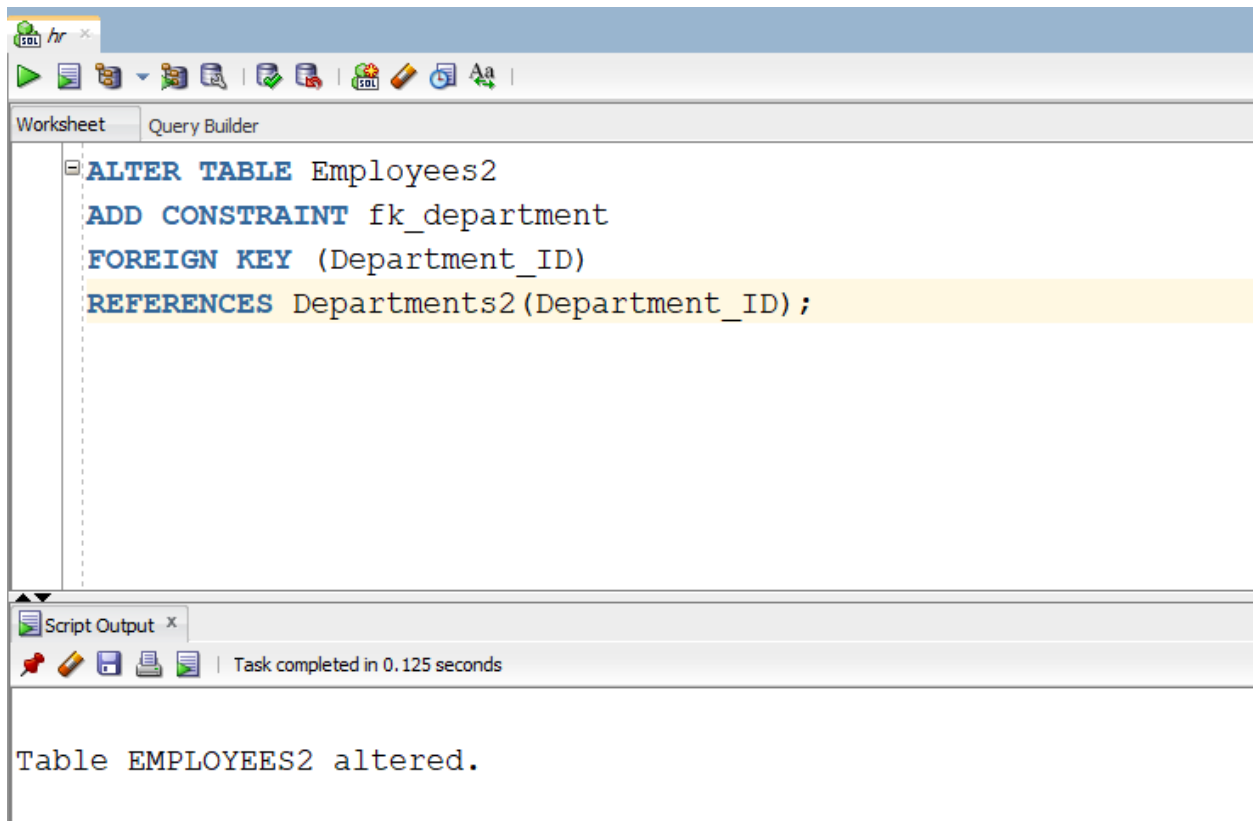
TASK 3:

The screenshot displays a SQL development tool interface. At the top, a tab labeled 'hr' is active. Below the toolbar, the 'Query Builder' tab is selected, showing the following SQL command:

```
ALTER TABLE Employees2  
ADD Department_ID NUMBER;
```

Below the query editor, the 'Script Output' tab is active, displaying the message: 'Table EMPLOYEES2 altered.' Above this message, a status bar indicates 'Task completed in 0.199 seconds'.

TASK 4:

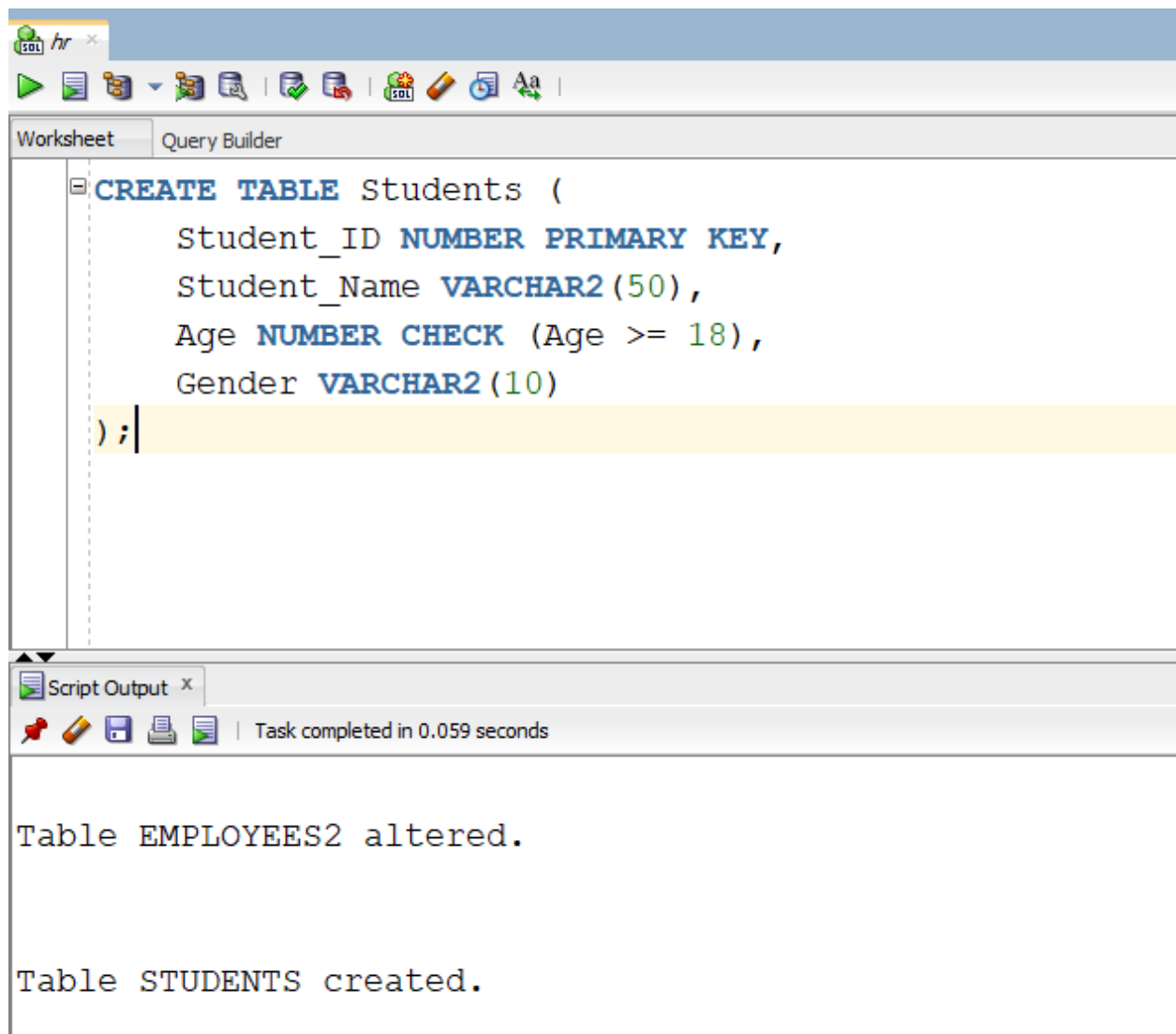


The screenshot displays the SQL Developer application window. The main editor area, titled 'Query Builder', contains the following SQL statement:

```
ALTER TABLE Employees2  
ADD CONSTRAINT fk_department  
FOREIGN KEY (Department_ID)  
REFERENCES Departments2 (Department_ID);
```

The 'Script Output' window at the bottom shows the result of the execution: 'Task completed in 0.125 seconds'. Below this, the message 'Table EMPLOYEES2 altered.' is displayed.

TASK 5:



The screenshot shows an SQL IDE interface. The top toolbar includes icons for running queries, saving, and other standard database operations. The main window is titled "Worksheet" and "Query Builder". It contains a SQL script to create a table named "Students". The script is as follows:

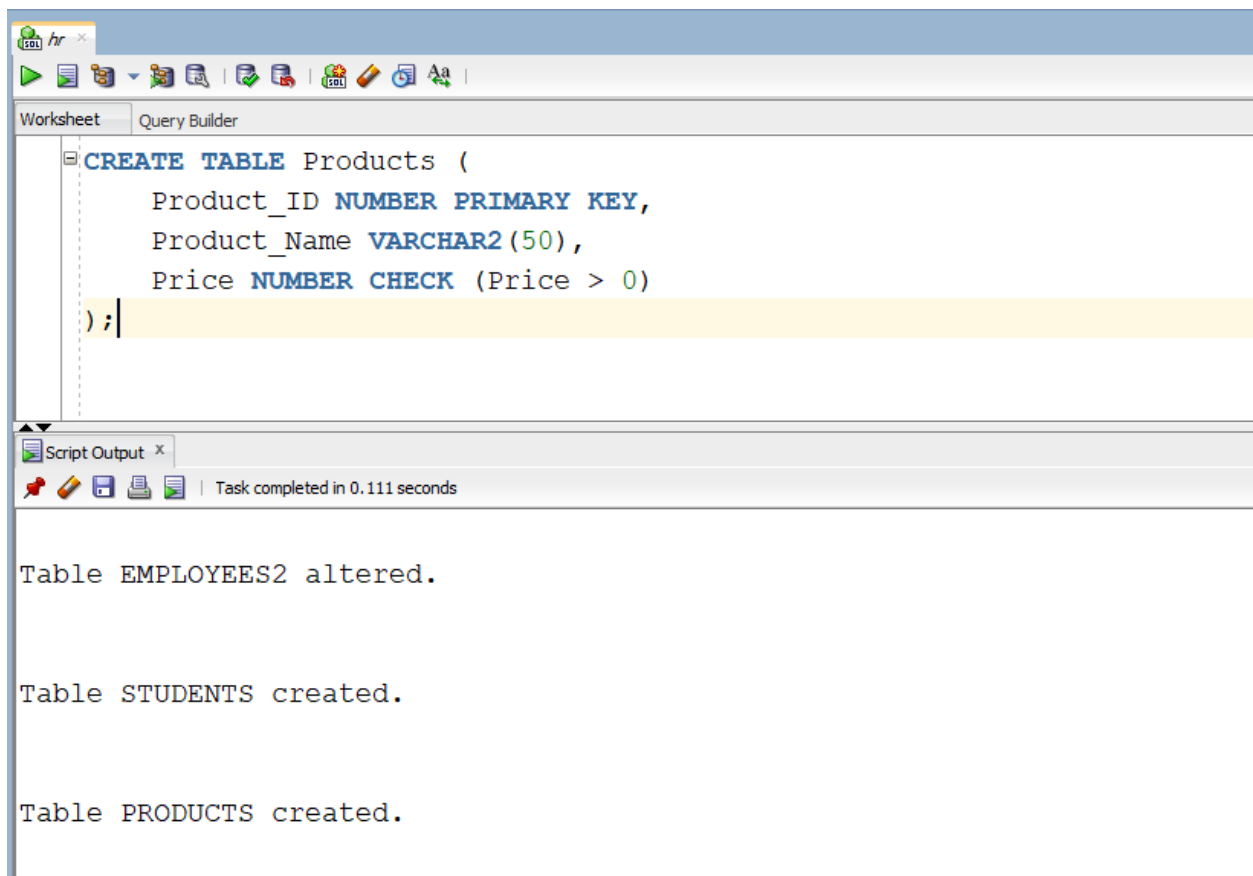
```
CREATE TABLE Students (  
    Student_ID NUMBER PRIMARY KEY,  
    Student_Name VARCHAR2(50),  
    Age NUMBER CHECK (Age >= 18),  
    Gender VARCHAR2(10)  
);
```

Below the script, the "Script Output" window shows the results of the execution. It indicates that the table "EMPLOYEES2" was altered and the table "STUDENTS" was created. The task was completed in 0.059 seconds.

Table EMPLOYEES2 altered.

Table STUDENTS created.

TASK 6:



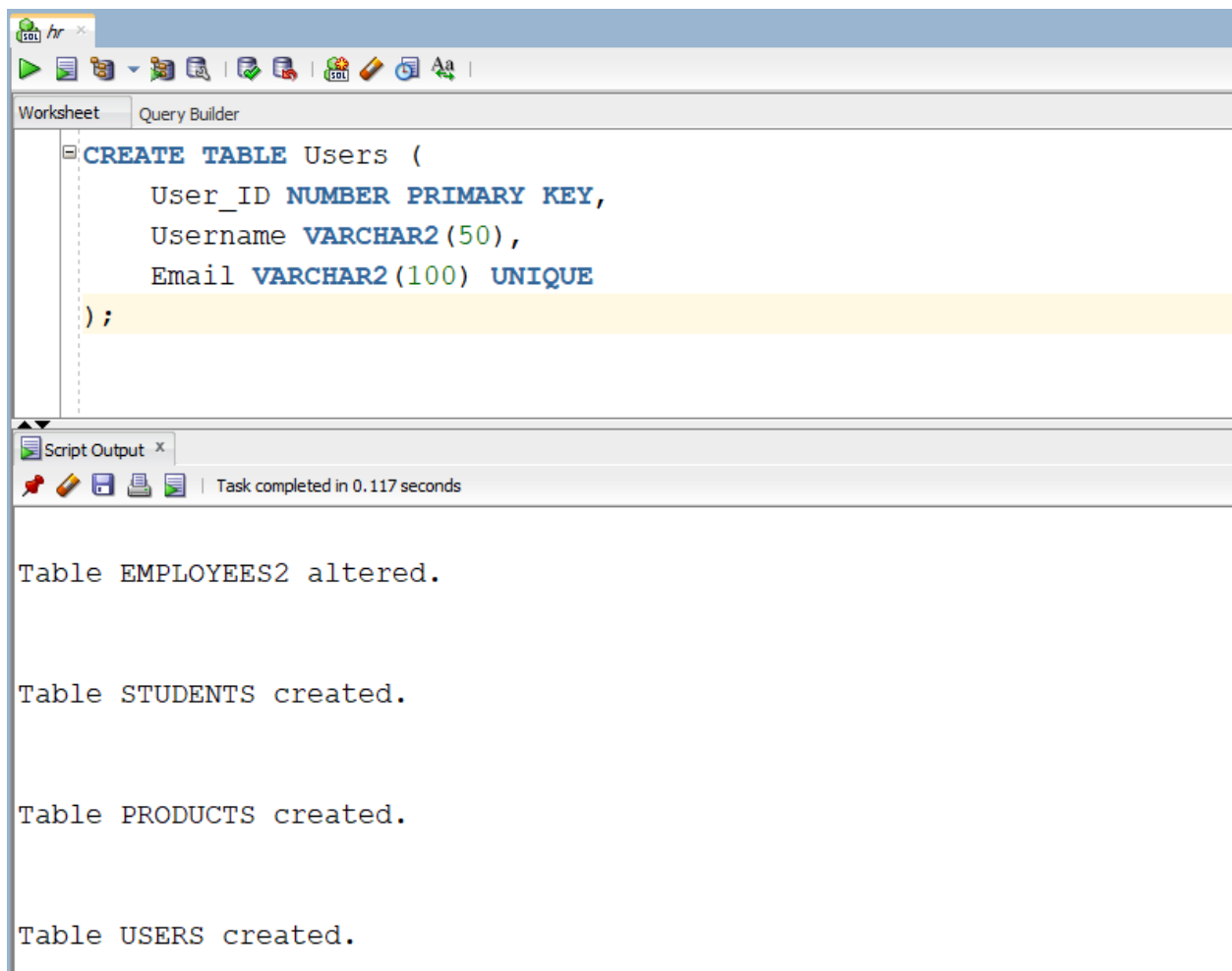
The screenshot shows a database query editor window with a toolbar at the top. The main area is divided into two panes: 'Worksheet' and 'Query Builder'. The 'Worksheet' pane contains a SQL script to create a table named 'Products'. The script is as follows:

```
CREATE TABLE Products (  
    Product_ID NUMBER PRIMARY KEY,  
    Product_Name VARCHAR2(50),  
    Price NUMBER CHECK (Price > 0)  
);
```

The 'Query Builder' pane shows the results of the script execution. The output is displayed in a 'Script Output' pane at the bottom, which indicates that the task was completed in 0.111 seconds. The output consists of three lines of text:

```
Table EMPLOYEES2 altered.  
  
Table STUDENTS created.  
  
Table PRODUCTS created.
```

TASK 7:



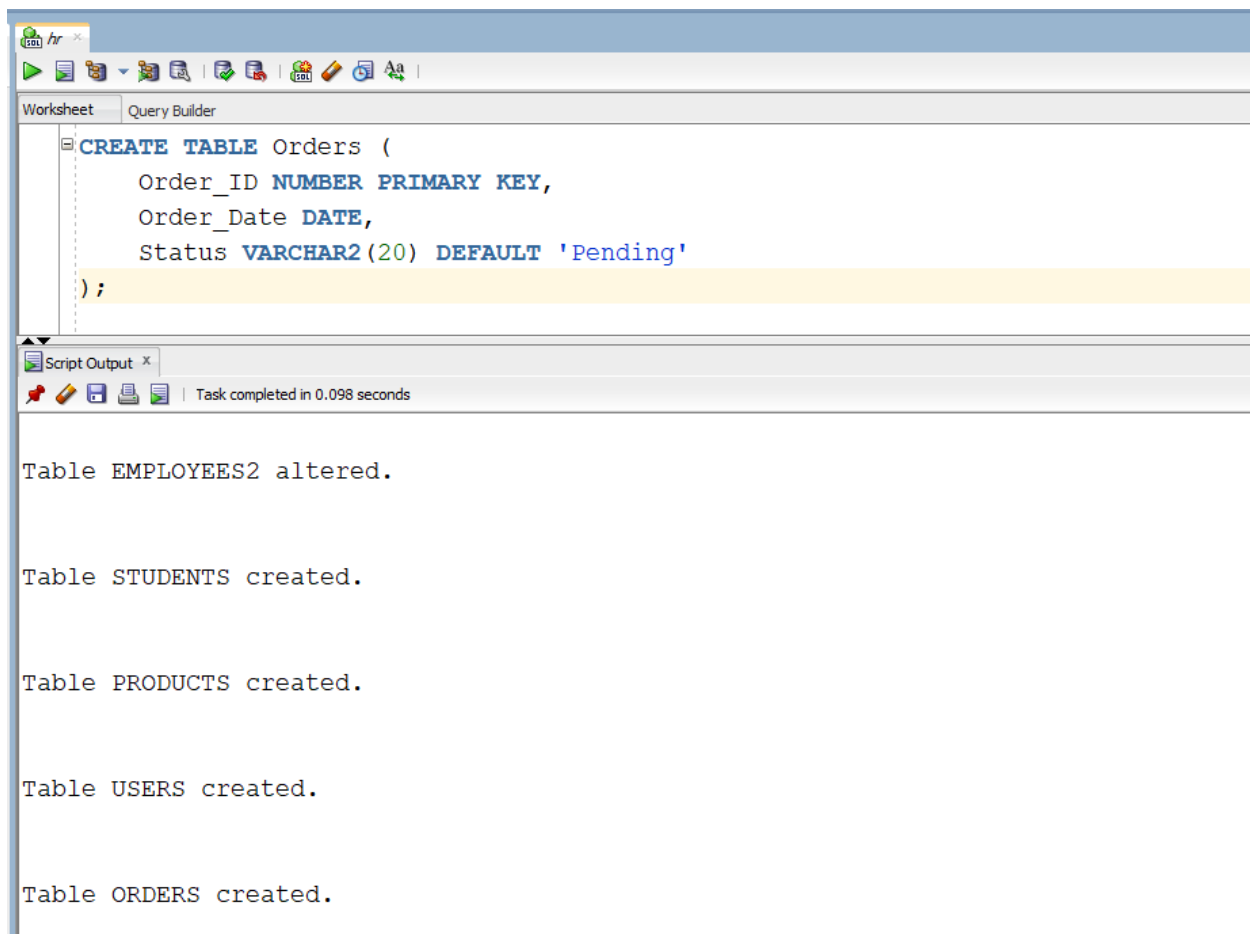
The screenshot displays an SQL IDE interface. The top toolbar includes icons for running queries, saving, and editing. The main workspace is titled 'Worksheet' and 'Query Builder'. It contains a SQL script to create a table named 'Users' with three columns: 'User_ID' (NUMBER, PRIMARY KEY), 'Username' (VARCHAR2(50)), and 'Email' (VARCHAR2(100), UNIQUE). The script is as follows:

```
CREATE TABLE Users (  
    User_ID NUMBER PRIMARY KEY,  
    Username VARCHAR2(50),  
    Email VARCHAR2(100) UNIQUE  
);
```

Below the script, the 'Script Output' window shows the results of the execution. It indicates that the task was completed in 0.117 seconds and lists the following messages:

- Table EMPLOYEES2 altered.
- Table STUDENTS created.
- Table PRODUCTS created.
- Table USERS created.

TASK 8:



The screenshot displays the Oracle SQL Developer environment. The top toolbar includes icons for running, saving, and editing. The 'Worksheet' tab is active, showing a SQL script in the 'Query Builder' view. The script is a SQL statement to create a table named 'Orders' with three columns: 'Order_ID' (NUMBER, PRIMARY KEY), 'Order_Date' (DATE), and 'Status' (VARCHAR2(20) with a default value of 'Pending'). The script is highlighted in yellow. Below the script, the 'Script Output' tab is visible, showing the execution results. The output consists of five lines of text: 'Table EMPLOYEES2 altered.', 'Table STUDENTS created.', 'Table PRODUCTS created.', 'Table USERS created.', and 'Table ORDERS created.' The status bar at the bottom indicates 'Task completed in 0.098 seconds'.

```
CREATE TABLE Orders (  
    Order_ID NUMBER PRIMARY KEY,  
    Order_Date DATE,  
    Status VARCHAR2(20) DEFAULT 'Pending'  
);
```

Table EMPLOYEES2 altered.

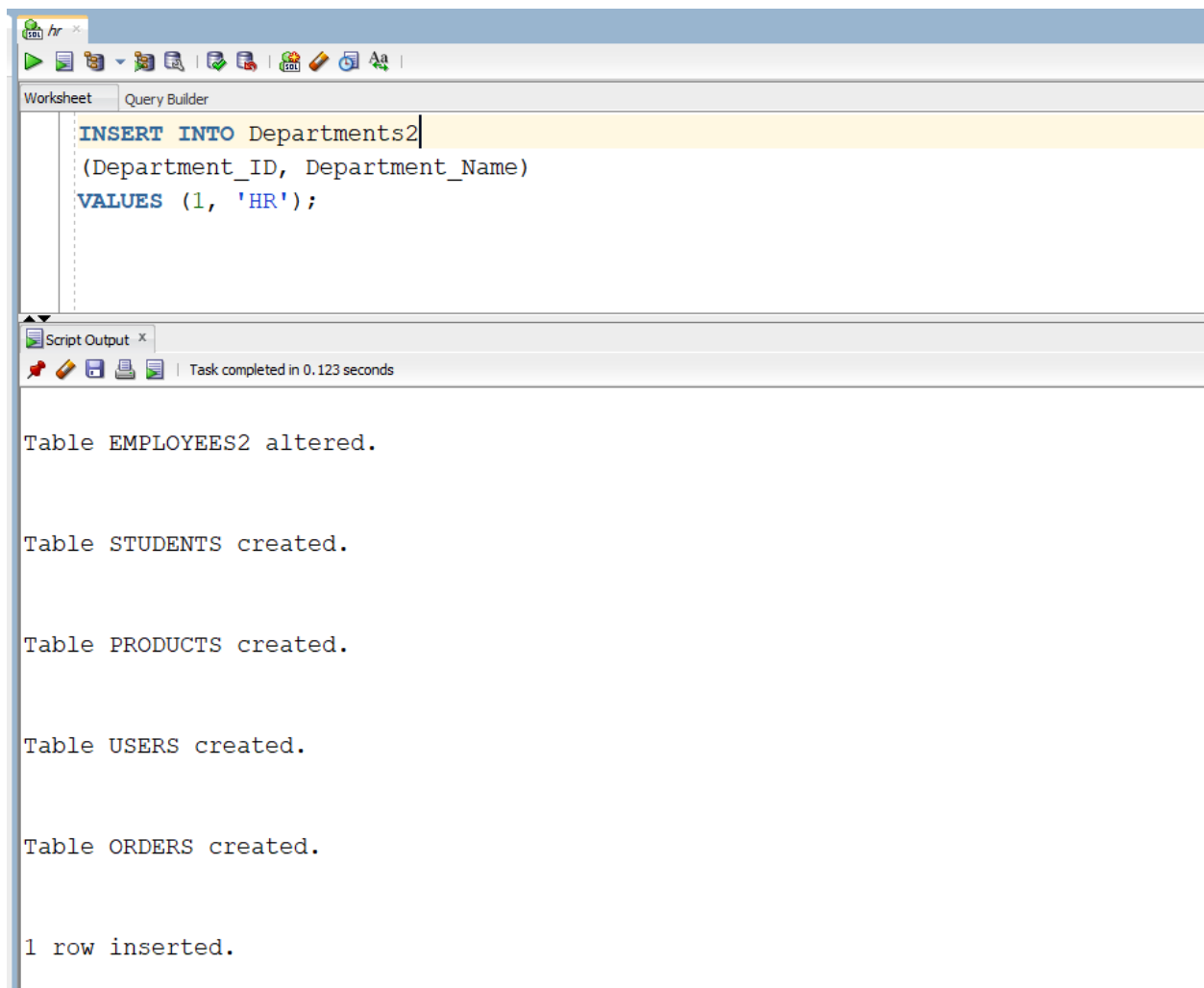
Table STUDENTS created.

Table PRODUCTS created.

Table USERS created.

Table ORDERS created.

TASK 9:



The screenshot displays a database management interface. At the top, a toolbar contains various icons for file operations and execution. Below the toolbar, a tab labeled 'Query Builder' is active, showing an SQL statement in a text editor. The statement is an INSERT command into a table named 'Departments2', specifying columns 'Department_ID' and 'Department_Name', and inserting a single row with values (1, 'HR'). Below the query editor, a 'Script Output' window is open, displaying the results of the query execution. The output consists of several status messages: 'Table EMPLOYEES2 altered.', 'Table STUDENTS created.', 'Table PRODUCTS created.', 'Table USERS created.', 'Table ORDERS created.', and '1 row inserted.'

```
INSERT INTO Departments2
(Department_ID, Department_Name)
VALUES (1, 'HR');
```

Script Output x

Task completed in 0.123 seconds

Table EMPLOYEES2 altered.

Table STUDENTS created.

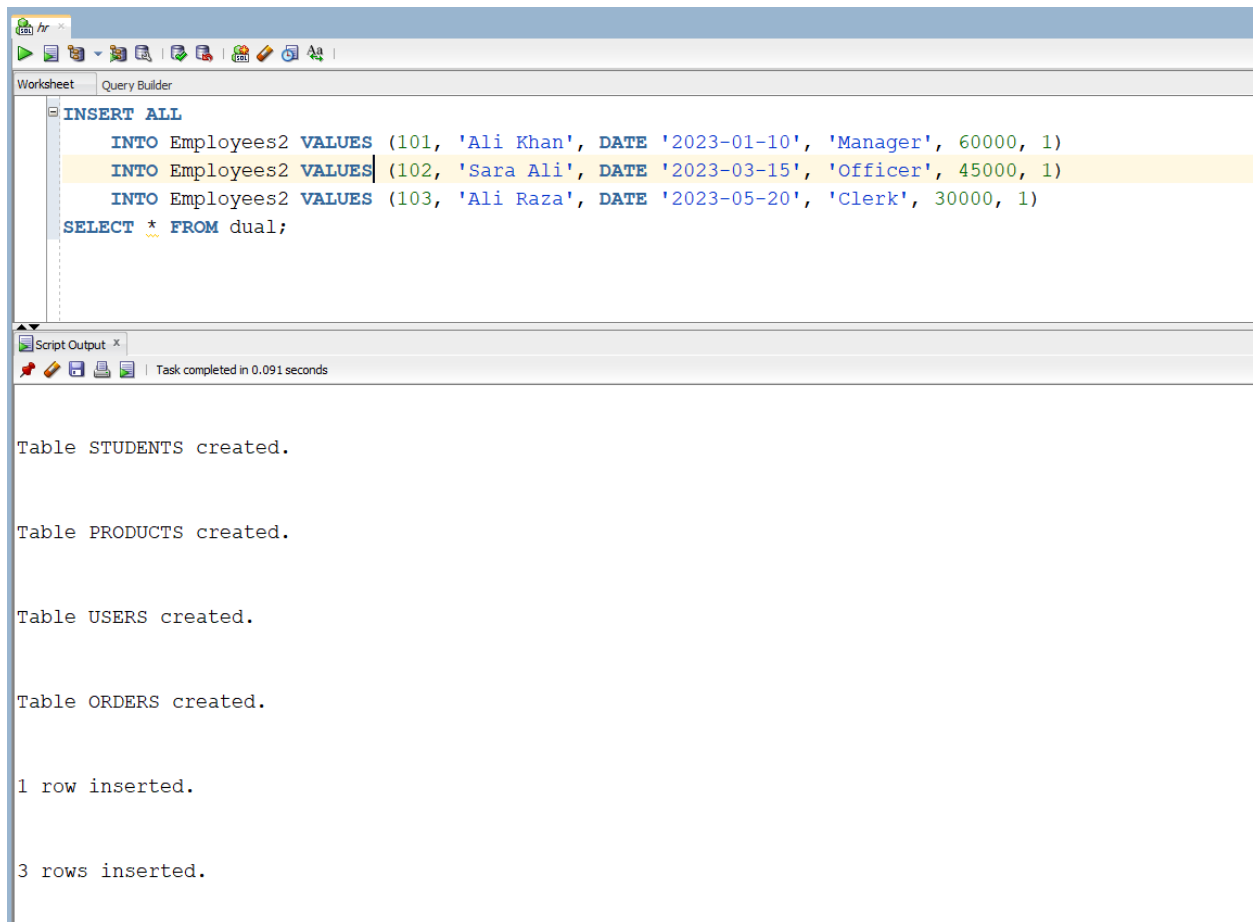
Table PRODUCTS created.

Table USERS created.

Table ORDERS created.

1 row inserted.

TASK 10:



The screenshot displays a database query editor interface. The top section, labeled 'Query Builder', contains the following SQL code:

```
INSERT ALL
  INTO Employees2 VALUES (101, 'Ali Khan', DATE '2023-01-10', 'Manager', 60000, 1)
  INTO Employees2 VALUES (102, 'Sara Ali', DATE '2023-03-15', 'Officer', 45000, 1)
  INTO Employees2 VALUES (103, 'Ali Raza', DATE '2023-05-20', 'Clerk', 30000, 1)
SELECT * FROM dual;
```

The bottom section, labeled 'Script Output', shows the results of the query execution:

```
Table STUDENTS created.

Table PRODUCTS created.

Table USERS created.

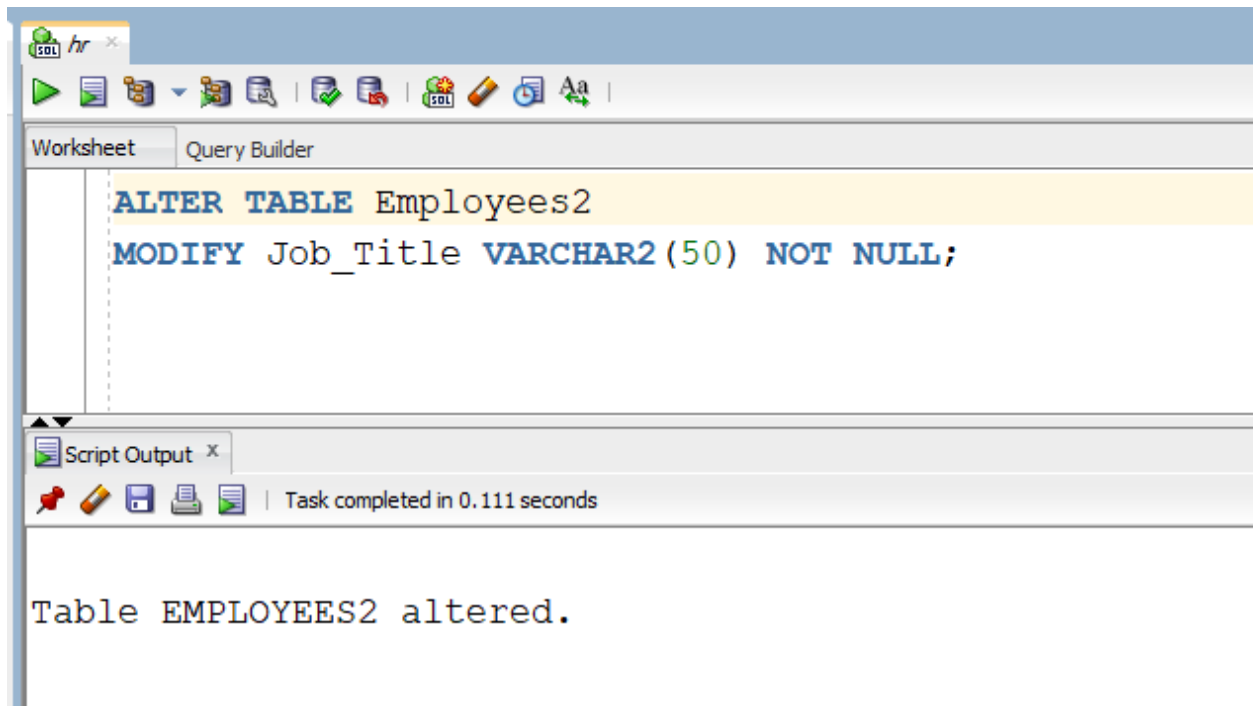
Table ORDERS created.

1 row inserted.

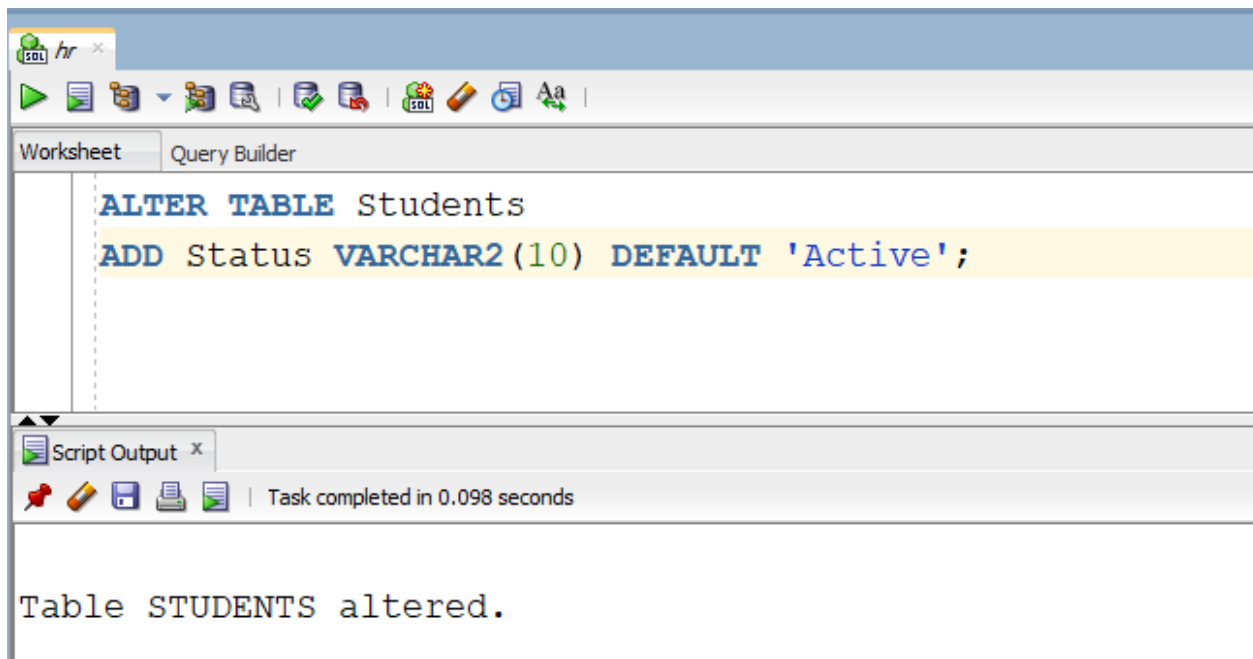
3 rows inserted.
```

POST-LAB TASK

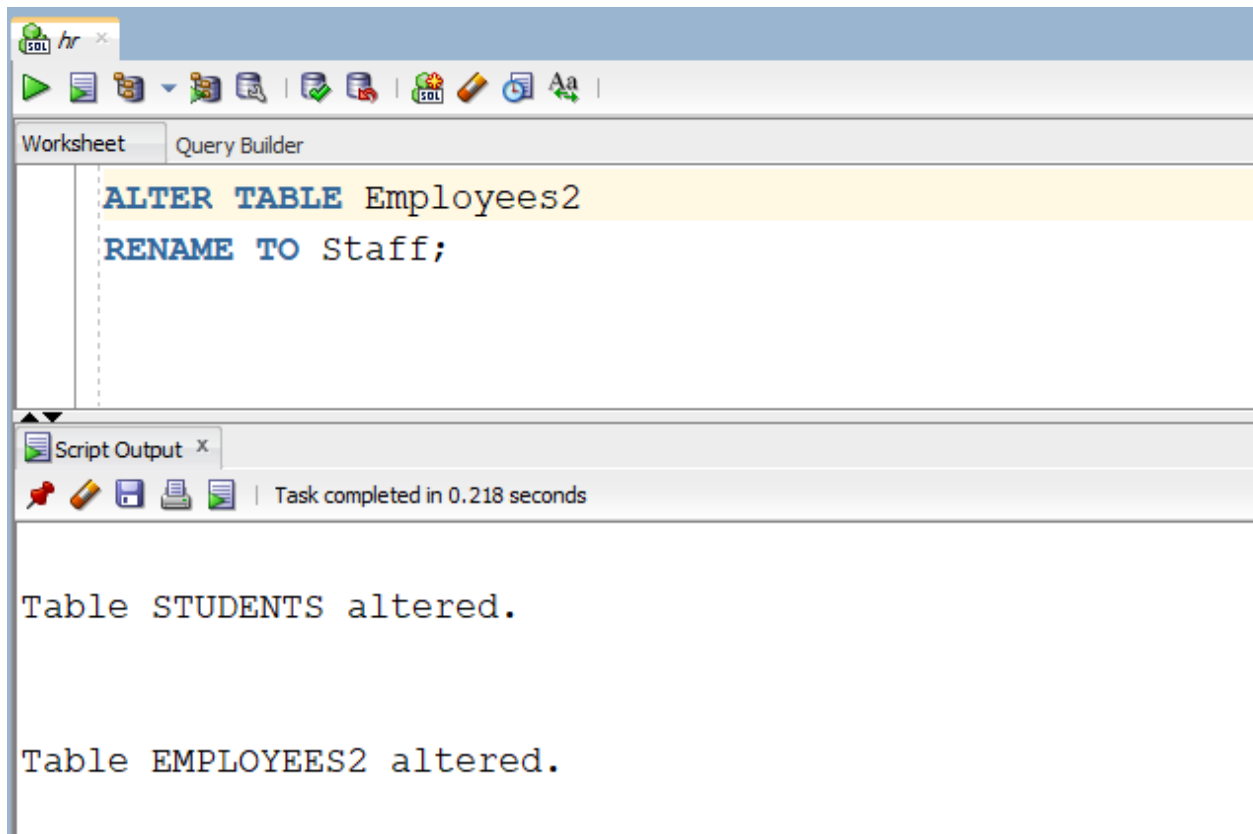
TASK 11:



TASK 12:



TASK 13:



The screenshot shows a SQL IDE window with a tab labeled 'hr'. The main editor area contains the following SQL query:

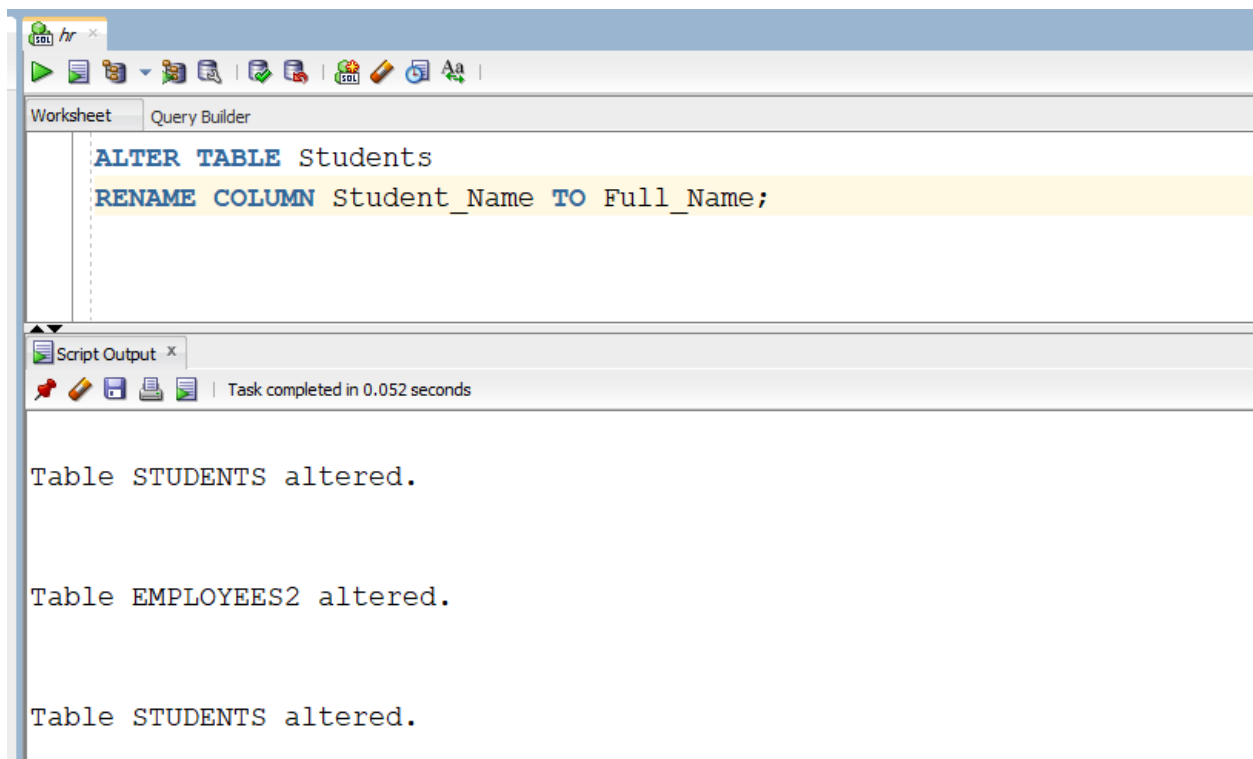
```
ALTER TABLE Employees2  
RENAME TO Staff;
```

Below the editor is a 'Script Output' window. It displays the results of the query execution:

```
Table STUDENTS altered.  
  
Table EMPLOYEES2 altered.
```

The 'Script Output' window also shows a status bar indicating 'Task completed in 0.218 seconds'.

TASK 14:



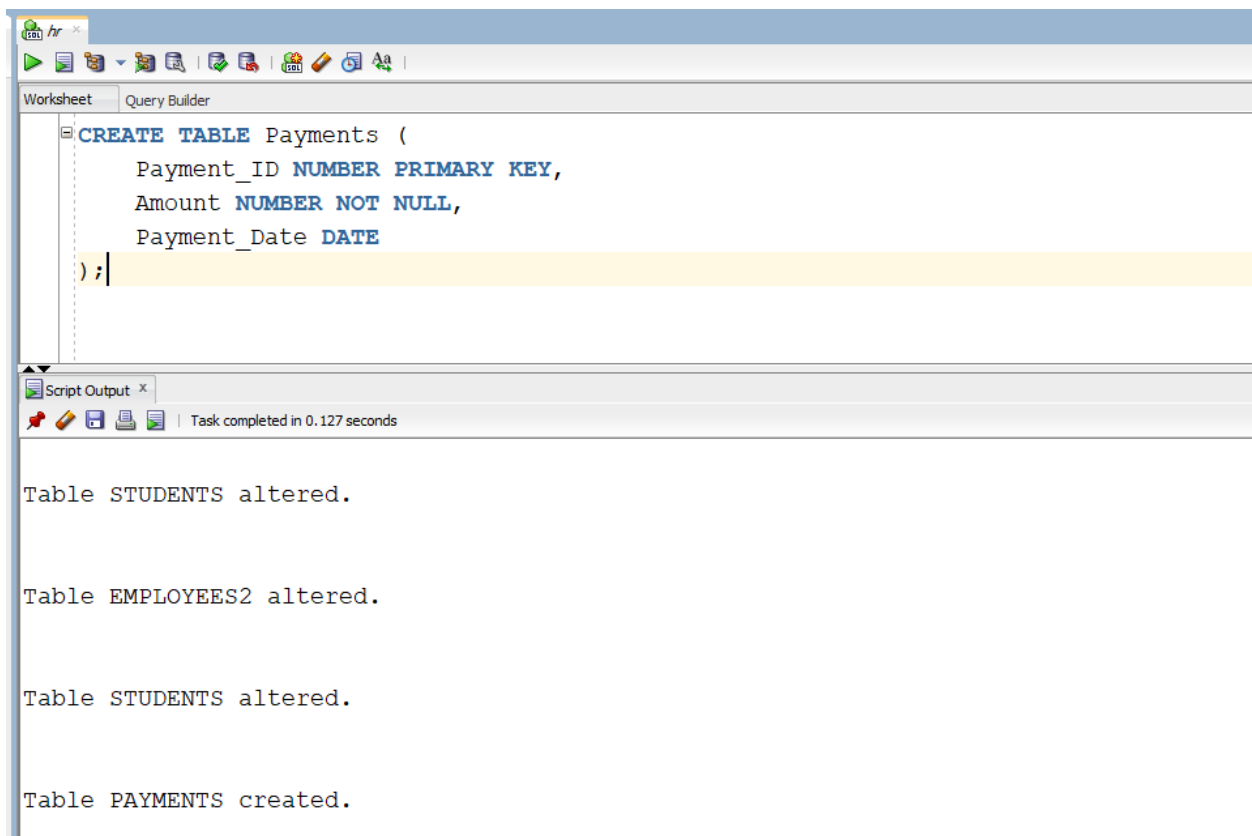
The screenshot displays the SQL Developer interface. The top toolbar includes icons for running queries, saving, and other database functions. The 'Query Builder' tab is active, showing the following SQL statement:

```
ALTER TABLE Students  
RENAME COLUMN Student_Name TO Full_Name;
```

Below the query editor, the 'Script Output' window shows the results of the execution:

```
Table STUDENTS altered.  
  
Table EMPLOYEES2 altered.  
  
Table STUDENTS altered.
```

TASK 15:



The screenshot displays the Oracle SQL Developer environment. The top toolbar includes icons for running, saving, and other database operations. The 'Query Builder' tab is active, showing a SQL script to create a table named 'Payments'. The script is as follows:

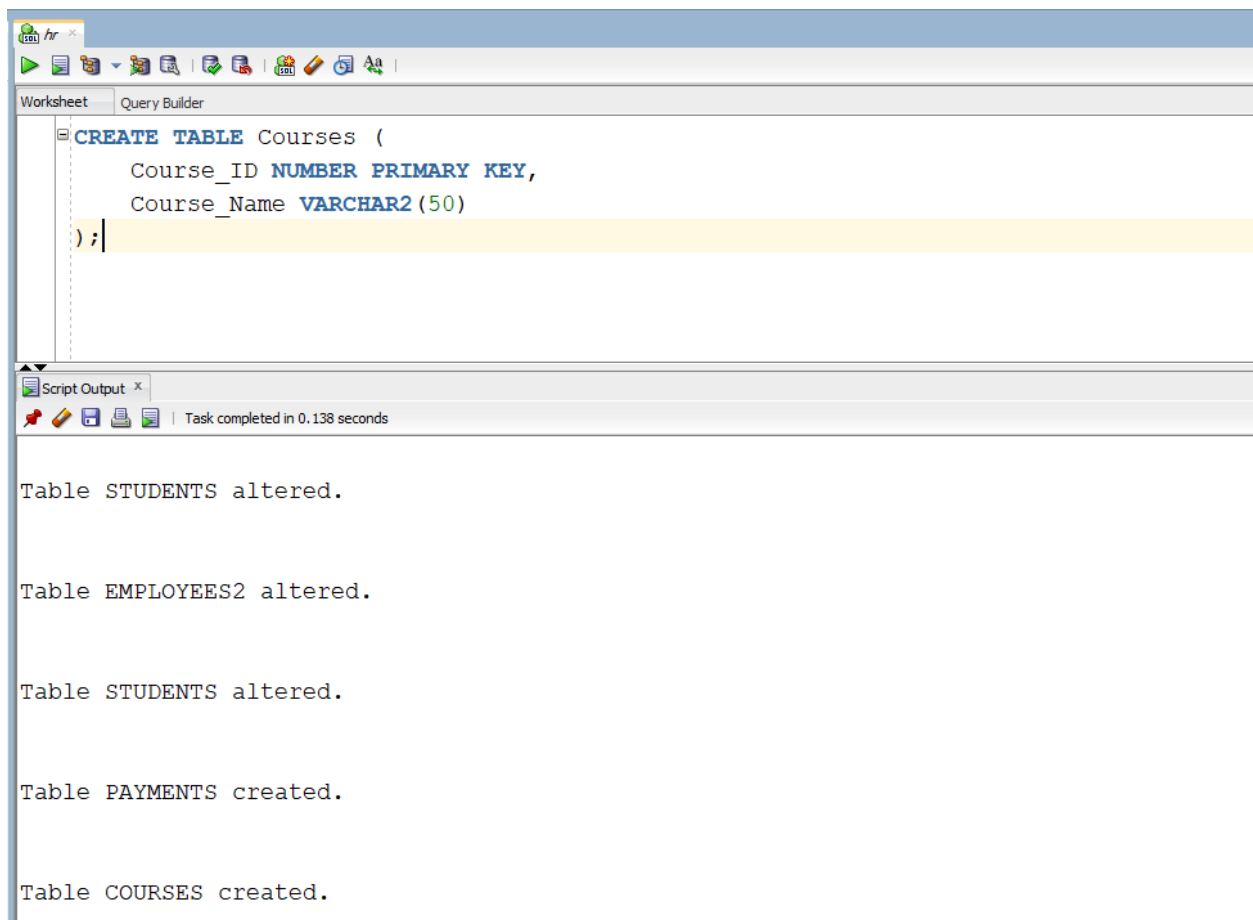
```
CREATE TABLE Payments (  
    Payment_ID NUMBER PRIMARY KEY,  
    Amount NUMBER NOT NULL,  
    Payment_Date DATE  
);
```

Below the script editor, the 'Script Output' window shows the results of the execution. It contains four messages:

- Table STUDENTS altered.
- Table EMPLOYEES2 altered.
- Table STUDENTS altered.
- Table PAYMENTS created.

The status bar at the bottom of the Script Output window indicates 'Task completed in 0.127 seconds'.

TASK 16:



The screenshot shows a database query editor interface. The top toolbar includes icons for running queries, saving, and other standard database operations. The main window is titled "Query Builder" and contains a SQL script. The script is as follows:

```
CREATE TABLE Courses (  
    Course_ID NUMBER PRIMARY KEY,  
    Course_Name VARCHAR2 (50)  
);
```

Below the script, the "Script Output" tab is active, displaying the results of the execution. The output consists of five lines, each indicating a table operation:

```
Table STUDENTS altered.  
  
Table EMPLOYEES2 altered.  
  
Table STUDENTS altered.  
  
Table PAYMENTS created.  
  
Table COURSES created.
```

The output also includes a status bar at the bottom indicating "Task completed in 0.138 seconds".

TASK 17:

The screenshot shows a database query editor with a 'Query Builder' tab. The SQL code in the editor is:

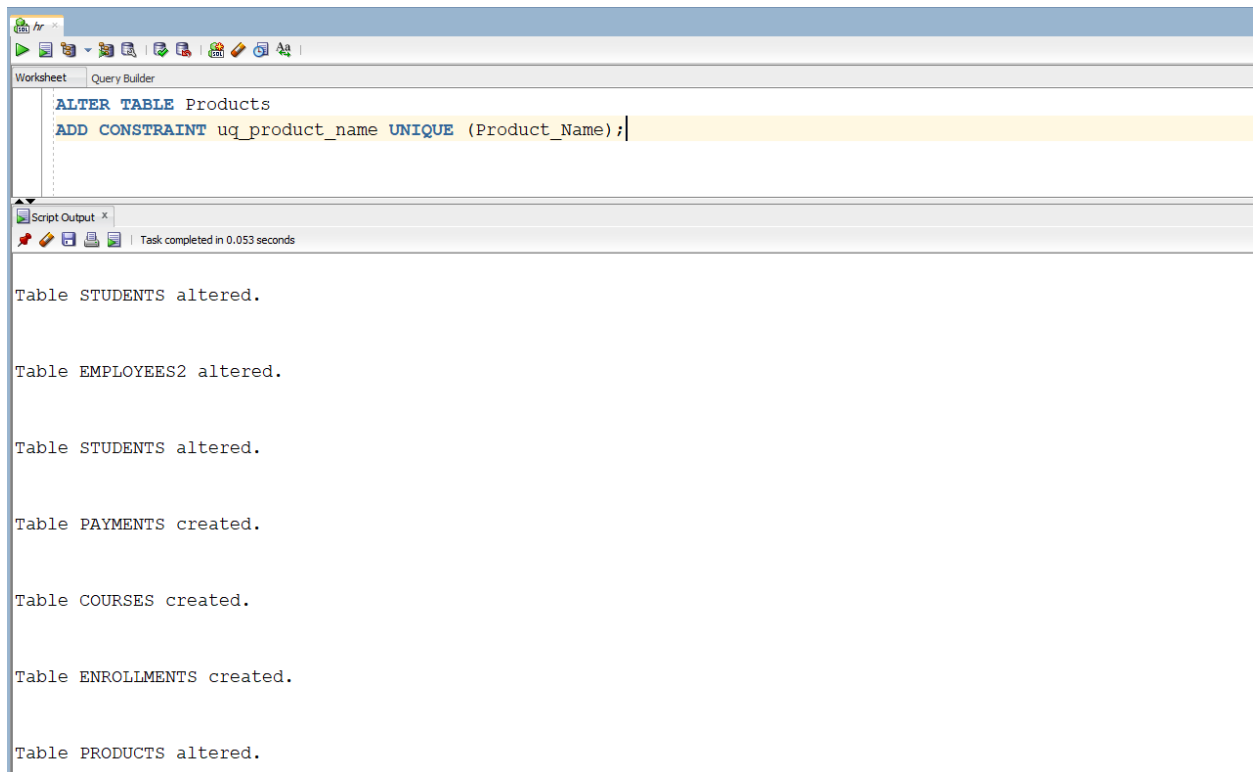
```
CREATE TABLE Enrollments (  
    Enrollment_ID NUMBER PRIMARY KEY,  
    Student_ID NUMBER,  
    Course_ID NUMBER,  
    FOREIGN KEY (Student_ID) REFERENCES Students(Student_ID),  
    FOREIGN KEY (Course_ID) REFERENCES Courses(Course_ID)  
);
```

Below the query editor, a 'Script Output' window displays the results of a script execution:

```
Table STUDENTS altered.  
  
Table EMPLOYEES2 altered.  
  
Table STUDENTS altered.  
  
Table PAYMENTS created.  
  
Table COURSES created.  
  
Table ENROLLMENTS created.
```

The 'Script Output' window also shows a status bar indicating 'Task completed in 0.105 seconds'.

TASK 18:



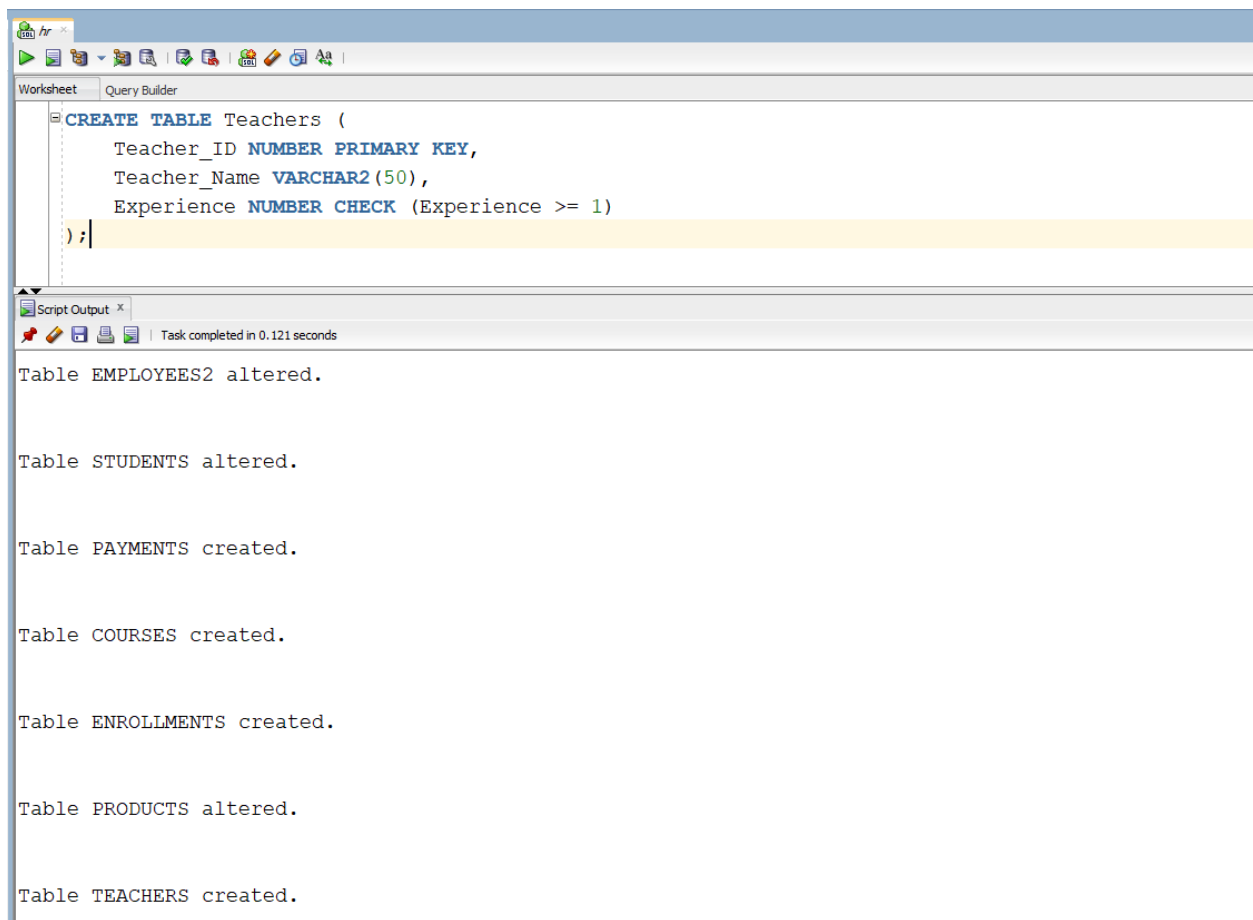
The screenshot displays a database management interface with a 'Query Builder' tab. The SQL script in the editor is:

```
ALTER TABLE Products  
ADD CONSTRAINT uq_product_name UNIQUE (Product_Name);
```

Below the script, the 'Script Output' window shows the execution results:

```
Table STUDENTS altered.  
  
Table EMPLOYEES2 altered.  
  
Table STUDENTS altered.  
  
Table PAYMENTS created.  
  
Table COURSES created.  
  
Table ENROLLMENTS created.  
  
Table PRODUCTS altered.
```

TASK 19:



The screenshot displays the Oracle SQL Developer environment. The top toolbar includes icons for running, saving, and other database operations. The 'Worksheet' tab is active, showing a SQL script in the 'Query Builder' area. The script is a CREATE TABLE statement for a table named 'Teachers'. The table has three columns: 'Teacher_ID' (NUMBER PRIMARY KEY), 'Teacher_Name' (VARCHAR2(50)), and 'Experience' (NUMBER CHECK (Experience >= 1)). The script ends with a semicolon. Below the script, the 'Script Output' window shows the results of the execution. It indicates that the task was completed in 0.121 seconds and lists the following messages: 'Table EMPLOYEES2 altered.', 'Table STUDENTS altered.', 'Table PAYMENTS created.', 'Table COURSES created.', 'Table ENROLLMENTS created.', 'Table PRODUCTS altered.', and 'Table TEACHERS created.'

```
CREATE TABLE Teachers (  
    Teacher_ID NUMBER PRIMARY KEY,  
    Teacher_Name VARCHAR2(50),  
    Experience NUMBER CHECK (Experience >= 1)  
);
```

Script Output x

Task completed in 0.121 seconds

Table EMPLOYEES2 altered.

Table STUDENTS altered.

Table PAYMENTS created.

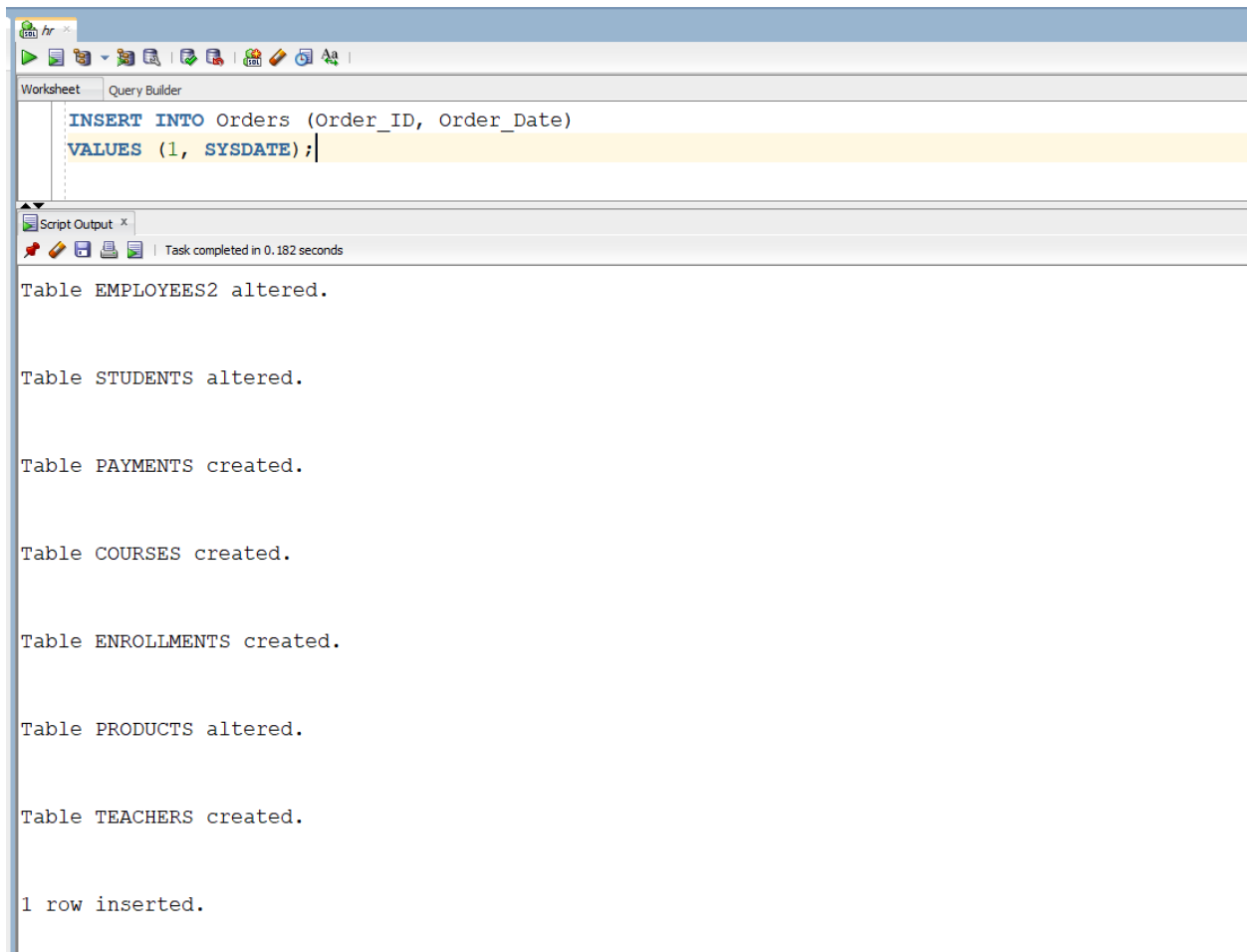
Table COURSES created.

Table ENROLLMENTS created.

Table PRODUCTS altered.

Table TEACHERS created.

TASK 20:



The screenshot shows the SQL Server Enterprise Manager interface. The 'Query Builder' tab is active, displaying the following SQL statement:

```
INSERT INTO Orders (Order_ID, Order_Date)
VALUES (1, SYSDATE);
```

The 'Script Output' tab is also visible, showing the execution results:

```
Table EMPLOYEES2 altered.

Table STUDENTS altered.

Table PAYMENTS created.

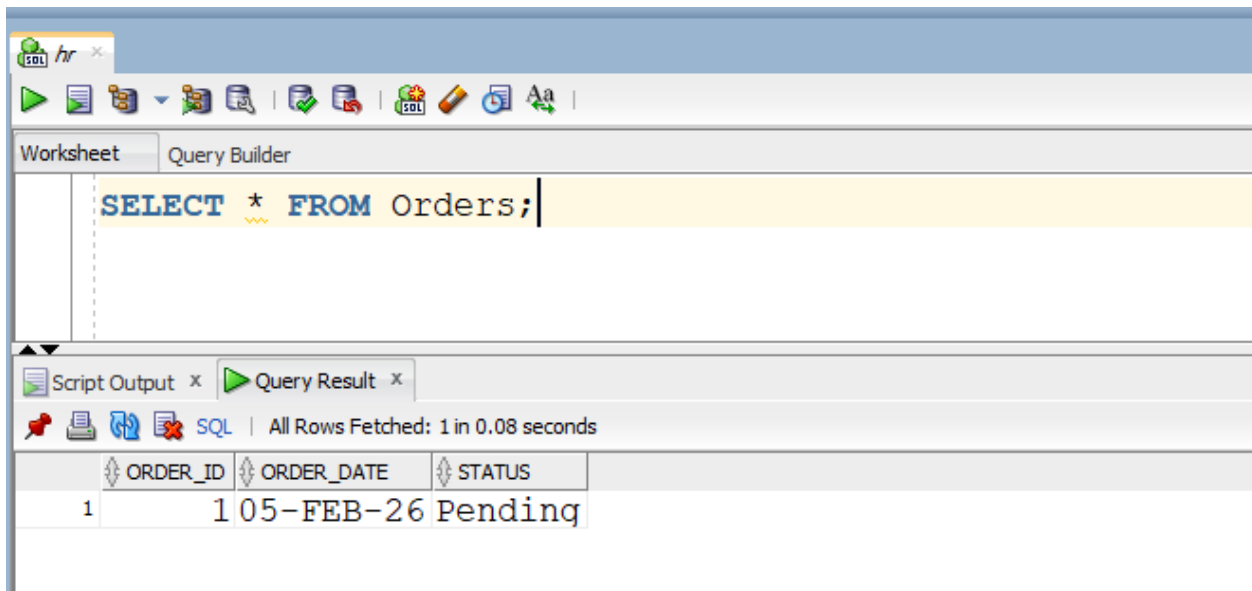
Table COURSES created.

Table ENROLLMENTS created.

Table PRODUCTS altered.

Table TEACHERS created.

1 row inserted.
```



The screenshot shows the SQL Server Enterprise Manager interface. The 'Query Builder' tab is active, displaying the following SQL statement:

```
SELECT * FROM Orders;
```

The 'Query Result' tab is also visible, showing the execution results:

```
All Rows Fetched: 1 in 0.08 seconds
```

	ORDER_ID	ORDER_DATE	STATUS
1	1	05-FEB-26	Pending